

**Written examination paper for SKY2100 – Cloud Security
Kristiania University College
Spring 2026**

1. Overview about exam and formal requirements

Examination paper released: 13.02.2026

Examination deadline: 20.02.2026

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Exam format: Individual written home examination

Final report format: A4 format, either LaTeX or Word format with the Arial font in size 11 and 1.0 line spacing. The limit is 12 pages max. This page limit includes text, screenshots or other figures and tables. The list of the used bibliography can come in addition to this and needs to be included on the last page of the submitted report. You can submit the final report as .docx or PDF file.

Support materials: Support materials allowed including the course books, course slides and course practical tasks. Therefore, you are not allowed to use search engines or any other material than stated above.

Grading scale: The Norwegian grading system uses the graded scale A - F, where A is the best grade, E is the lowest pass grade, and F is failed.

Weighting: 100% of the overall grade

Plagiarism control: We expect your independent work. Use citations and quotations if there is material you want to include in the report. This includes text and other material e.g. like figures, tables. Use APA7 for referencing from the course source. Also, only use intext citations. This means, cite at the exact place in a sentence whenever you quote or take information from a source. Information: When it comes to citation of the books, reference the page(s). For the slides, refer to which lecture (1-12) and which slide number (number on lower right side of a slide, not the page number which is shown in your PDF reader). References without these indicators are considered plagiarism.

Tasks format: Answering task A is mandatory. Then, you can choose practical task B OR answer task C, in case your PC has some issues with virtualization. If you choose task C, please also include the screenshot of the error or system message (with time and data visible on the screenshot) showing that you cannot perform the virtualization task.

Screenshots: Your "candidate number" is mandatory to be part of the screenshots for documentation. You can do that e.g. by changing your username in the virtual machine to your candidate number or when making screenshots opening an editor with your candidate number and include it in the screenshot. Provide only screenshots were asked for in the tasks.

Learning outcomes as per course description:

Knowledge. The student...

- can define and explain how cloud services work
- can explain the terms Software as a Service, Platform as a Service and Security as a Service
- can define and explain how cloud solutions are secured and what differences and similarities this is in relation to internal servers
- can explain the difference between a Cloud Native application and a cloud migration that uses the Lift and Shift methodology
- can define and explain the role of a Web Application Firewall and how a WAF can be part of a security architecture

- can define and explain the development methodology DevSecOps and how this differs from traditional product development

Skills. The student ...

- can use, configure and manage security in a cloud solution
- can install and configure a Web Application Firewall and use it to protect a cloud solution
- can use security and vulnerability analysis tools integrated with the cloud solution
- can use integrated logging in cloud solutions and understand how these logs can be integrated with a SIEM tool

General competence. The student can...

- reflect the architectural requirements and give advice about security architecture and implementation of security solutions in cloud services
- describe and define the advantages and disadvantages of establishing or moving a project / service to the cloud from a security perspective

2. Exam Task

Task A (70% of the grade)

After graduating from Kristiania, you are hired as a Junior Cloud Security Architect at NordicRide, a fast-growing mobility-as-a-service (MaaS) company headquartered in Copenhagen, Denmark.

NordicRide operates a digital platform that integrates e-scooters, e-bikes, car-sharing, and public transport ticketing into a single mobile application. The company currently employs approximately 300 people, with offices in Copenhagen, Oslo, Helsinki, and Berlin.

Until recently, NordicRide relied on a traditional on-premise infrastructure hosted by a third-party data center provider in Denmark. Due to rapid growth, increasing mobile usage, and expansion into new European and US markets, the company has decided to migrate its platform to the cloud.

The platform processes real-time location data, payment and transaction data, personal user profiles as well as sensitive behavioral data for route optimization and dynamic pricing

Security, availability, and regulatory compliance are critical, especially during peak usage hours (rush hours, large events, and bad weather conditions).

- **Task A.1:** Use our main course books and inline referencing to explain which Service and Deployment Model you would recommend in future. What trade-offs can you imagine in this scenario based on our course books and slides and why?
- **Task A.2:** Describe the data protection and privacy requirements from our main course books and related Azure sources that could impact NordicRides expansion into new regions. Focus on how these regulations affect cloud operations, especially concerning mobile data and social media integrations. Additionally, provide 3–5 recommendations for maintaining secure data handling and ensuring regulatory compliance when scaling NordicRides cloud infrastructure, based on our course books and slides.
- **Task A.3:** Imagine you are tasked with creating a tailored IT security training for new hires, focusing on developers and software architects working in NordicRides fast-paced fashion e-commerce environment. This training should address best practices for securing applications that handle real-time transactions, customer data, and social media integrations. Outline the key security steps you would recommend throughout the software development lifecycle (SDLC), with a focus on protecting sensitive data in mobile apps and during high-traffic events like flash sales. Additionally, recommend a security framework that fits this dynamic, mobile-driven industry and explain your choice based on concepts from our course books and lecture slides.
- **Task A.4:** Imagine you are part of NordicRides regular IT security training for new hires, focusing on developers and software architects in the fast-paced fashion e-commerce environment. This training should cover best practices for securing applications that handle real-time transactions, customer data, and social media integrations. Outline the key security steps you would recommend based on the current version of the NIST CSF, with a focus on protecting sensitive data in mobile apps and during high-traffic events like flash sales. Explain your choice using each part of the current version of the NIST CSF from our course books and lecture slides.
- **Task A.5:** Include screenshot of the completed training modules in Azure <https://learn.microsoft.com/en-us/users/> by clicking on "Training". To change your username for

anonymization reasons, follow <https://learn.microsoft.com/en-us/users/me/settings> where you can change the “Display name” and optional the “Username and URL” if needed.

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Important: For all tasks, use information from the slides and course books. Refer to the slide/page where you have found an information for your solution proposal in accordance with the formal requirements mentioned on the first page of the exam.

Marking Guidance:

- For Task A.1, students should clearly define and justify their choice of a cloud Service Model (IaaS, PaaS, SaaS) and Deployment Model (Public, Private, Hybrid, Multi-cloud) using concepts from the course books. They must discuss trade-offs, such as scalability vs. cost, security vs. flexibility, and vendor lock-in risks. Justifications must align with course content for higher grades using inline references.
- For Task A.2, students must outline key data protection and privacy regulations relevant to Trendora’s expansion. Responses should use sources from course materials and include 3–5 practical recommendations for ensuring compliance (e.g., data encryption, access controls, region-specific compliance). Justifications must align with course content for higher grades using inline references.
- For Task A.3, students must propose an IT security training tailored to developers and software architects, addressing real-time transactions, customer data, and social media security. The training should outline key security steps in the Software Development Lifecycle (SDLC) and recommend a security framework suited for a fast-moving e-commerce company. Justifications must align with course content for higher grades using inline references.
- For Task A.4, students should present a structured security training plan based on NIST CSF 2.0, covering its six core functions (Govern, Identify, Protect, Detect, Respond, Recover) with a clear link to securing mobile apps and handling high-traffic flash sales. Responses should demonstrate a clear understanding of NIST CSF principles and their application in e-commerce security. Justifications must align with course content for higher grades using inline references.
- For Task A.5, students must submit a screenshot of completed 23 Azure training modules with anonymized usernames. Missing or incomplete screenshots will result in lower marks.
- High-quality submissions provide clear explanations, strong justifications, and inline references from course materials. Weak answers lack depth, miss key concepts or fail to apply theories to Trendora’s scenario. Furthermore, weak answers include non-referencing.

Task B (30% of the grade)

- **Task B.1:** Install Ubuntu Virtual Machine either in VMware.
- **Task B.2:** In the command line, access information about a) the number of CPU/cores and bogomips per core, b) open network port and explain the output. What do you see and where can you identify the answers in the terminal output? **Important:** Make a screenshot of the results of a) and b) which includes your candidate number (e.g. by opening an editor next to your terminal and write your candidate number inside).
- **Task B.3:** Install the Nginx server in your VM. Open two terminal windows and, in one, perform a simulation of the DDoS attack: “`ab -n 123456 -c 120 127.0.0`”. In another, run the interactive “`top`” command. What was the highest CPU load you managed to achieve? Give a explanation. **Important:** Include the screenshot in the report showing the DDoS attack and the “`top`” command. The screenshot needs to include you candidate number (e.g. by opening an editor next to your terminal and write your candidate number inside).
- **Task B.4:** Install docker and run the same experiment in the docker container with Nginx. **Important:** Include a screenshot in the report showing your docker installation, test results, and system load and compare if there are any differences with the results from (B.3). The screenshot needs to include you candidate number.

- **Task B.5:** Analyse the average number of requests/concurrencies threads your VM machine can handle before it is offline (“DDoS”) due to too many requests. Check if your VM can hold one million requests against the web server: “ab -n 1234567 -c 2300 127.0.0.1”. What is the result? Give a detailed explanation.

Marking Guidance:

- For Task B.1, students must successfully install an Ubuntu Virtual Machine using VMware or VirtualBox. Submissions with a successful installation increase the grade and an incorrect setup will reduce the score.
- For Task B.2, students need to retrieve and explain system information related to CPU cores, bogomips, and open network ports. The explanation should correctly identify where to find these details in the terminal output. Screenshots must clearly show results alongside the candidate number. Missing explanations or unclear screenshots lower the grade.
- For Task B.3, students must install Nginx, simulate a DDoS attack using Apache Benchmark (ab), and monitor CPU usage via the top command. Higher scores go to students who provide a clear explanation of CPU load impact. Screenshots must clearly show results alongside the candidate number. Missing explanations or unclear screenshots lower the grade.
- For Task B.4, students should install Docker, run the same DDoS simulation inside a container, and compare the results with Task B.3. Stronger answers will highlight performance differences, explain system load variations, and include all required screenshots with their candidate number. Screenshots must clearly show results alongside the candidate number. Missing explanations or unclear screenshots lower the grade.
- For Task B.5, students must analyze how many concurrent requests their VM can handle before becoming unresponsive and attempt a one-million-request test. The explanation should discuss performance limits and why the VM either withstands or fails under heavy load. Justifications must align with course content for higher grades.
- Higher-quality submissions provide clear explanations, complete screenshots with candidate numbers, and proper analysis of results. Weak answers miss key technical details, lack screenshots, or fail to provide meaningful interpretations of system behavior.

Task C (30% of the grade)

A multinational financial services company, headquartered in Amsterdam (Netherlands), plans to implement a cloud-based platform for document management and regulatory reporting. The company operates globally, including offices in Norway, Germany, and the US. The platform must handle confidential client data, audit documentation and regulatory reports

- **Task C.1:** You work as an IT Consultant at this company and your task is to design a new cloud-based architecture for this platform. Describe in detail the infrastructure and platform architecture, including considerations for data protection, IT security, and IT compliance. Additionally, evaluate the financial factors associated with migrating to a cloud service. Also, consider cybersecurity protection mechanisms discussed in the lectures. Do you need security measures such as Identity and Access Management (IAM), encryption, or DDoS protection? Explain your choices based on key concepts from the course textbooks and lecture slides.

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Important: Only use information from the slides and course books. Refer to the slide/page where you have found an information for your solution proposal. Also, you can make assumptions like certain IT Security/Cloud/Compliance certifications and assurance reports in place or needed based on our course books and slides.

Marking Guidance:

- For Task C.1, students must design a detailed cloud-based platform tailored to the financial services company needs. The response should include the infrastructure and platform architecture, considering scalability, performance, and regulatory compliance. Justifications must align with course content for higher grades using inline references.
- Key data protection and IT security considerations should be discussed, including compliance within the US, EU (GDPR), and – optionally - Norwegian regulations. Good answers include explanations of needed security mechanisms such as Identity and Access Management (IAM), encryption, DDoS protection, MFA and secure API integrations.
- Students must also evaluate financial factors associated with cloud migration. These include cost-efficiency, operational expenses and scalability. Discussions on pay-as-you-go pricing, resource optimization, and cost trade-offs will result in higher grades.

4. AI Declaration

Copy this into your submission and fill it out. Mark with bold letters what apply (Yes/No):

Declaration of the Use of AI Tools and Resources

The checklist below must be completed:

- **Text correction:** Have parts of the text been checked using tools such as Grammarly, Ginger, LanguageTool, or similar? Yes / No
- **Text generation:** Have parts of the text been generated using tools such as ChatGPT, Jasper, WriteSonic, or similar? Yes / No
- **Writing assistance:** Have any ideas or approaches in the report been suggested by tools such as ChatGPT, Google Bard, or similar? Yes / No
- **Image generation:** Have any images/figures in the report been generated using tools such as Midjourney, DALL·E, or similar? Yes / No
- **Other AI tools:** Have you used any other AI tools? If yes, please specify here:

Have you used any AI-based tools or resources in the preparation of this report?

If yes, specify the tool and its purpose below.

1. **Text generation** – If you used text-based tools, which tasks were completed with AI assistance, and how?
2. **Images and figures** – If you used image/figure generators, which images or diagrams were AI-generated?

I declare that all use of AI tools or resources has been clearly disclosed either in the report or in this declaration form.

Candidate number / Date